Omar ALIBI

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Profile

- Electrical Engineering graduate from the National Engineering School of Tunis (ENIT), specializing in Advanced Reconfigurable and Real-Time Microelectronic Systems. Passionate about embedded systems, IoT, and connected technologies, with hands-on experience in PCB design, firmware development, and cloud-connected energy IoT solutions, including an end-of-studies internship in Malta. Driven by curiosity and innovation, seeking to contribute to challenging projects, build impactful technologies, and grow within dynamic engineering teams.

Technical Skills

Embedded Systems & Electronics

- Arduino, ESP32/8266, STM32, nRF52, Raspberry Pi, BeagleBone, Jetson Nano, Zybo Z7, Nordic PPK2, J-Link
- UART, SPI, I2C, CAN, Modbus, M-Bus (Wired), FreeRTOS, Embedded Linux
- Altium, Eagle PCB, LTSpice, PSpice, Keil, ModelSim, QuestaSim, Vivado
- C, C++, Rust, VHDL, Python

IoT, Cloud & Tools

- MQTT, WebSocket, ThingsBoard, Node-RED, Grafana, Firebase, Node.js, PySide6, Electron.js
- Git/GitHub, Docker, Linux, MATLAB

Professional Experience

End-of-Studies Engineering Intern, IoT Solutions Ltd *(Malta, April 2026 – August 2026)*

- **Electronics:** Designed an M-Bus master daughterboard around boost converter, then simulated, breadboard-tested, and laid it out in Altium.
- **Embedded Software:** Developed nRF52 firmware for wired M-Bus
- **Production Tooling:** Modernized the end-of-line tester as an asynchronous app with PPK2 current profiling, J-Link flashing, serial diagnostics, backend registration, and per-run failure logs.
- **Cloud & Analytics:** Built a unified ThingsBoard workspace for electricity and water, with custom widgets.

Engineering Intern, OnWireWay *(June 2025– July 2025)*

- Developed custom ESP8266 firmware for Sonoff smart switches to connect them to a private server (embedded web server, REST API, solar-time scheduling), and built a Flutter mobile app with Firebase for remote MQTT control.

Technical Intern, STEG *(July 2024)*

- Analyzed electrical installations and distribution networks; performed preventive maintenance and troubleshooting on industrial equipment.

Technical Projects

RISC-V System-on-Chip Implementation in VHDL

- Designed and implemented an RV32I SoC with AXI4, memory, UART/GPIO peripherals, and a custom coprocessor; verified in QuestaSim and deployed on Xilinx FPGA.

Real-Time DC Motor Control on BeagleBone Black

- Implemented real-time closed-loop DC motor speed control on embedded RT Linux with a custom kernel module, integrating ADC feedback, encoder measurement, PWM generation, and a PID controller.

Inertial Navigation of a Drone (IMU Fusion): PFA2

- Implemented an IMU with sensor fusion using EKF on an STM32 with a custom Qt/C++ simulation environment.

IoT Dashboard for Connected Machines

- Built a real-time industrial monitoring dashboard using Electron, MQTT, JWT auth, WhatsApp alerts (Twilio), and automated PDF reporting.

Multi-Threaded Real-Time Embedded System

- Designed a FreeRTOS STM32 system integrating OLED, USB HID, and ESP8266 with concurrent tasks for email, computation, and mini-games.

STM32 Digits Classifier

- Created a real-time handwritten digit recognition system using neural networks on STM32; draws digits in a desktop GUI with on-chip AI inference via X-CUBE-AI.

Education

National Engineering School of Tunis (ENIT) *(2023 – 2026)* – Engineering Degree in Electrical Engineering, Advanced Reconfigurable and Real-Time Microelectronic Systems

Preparatory Institute for Engineering Studies El Manar *(2021 – 2023)* – Admitted through the National Engineering Schools Entrance Examination 2023

Carthage Présidence High School *(2019 – 2021)* – Baccalaureate in Technical Sciences, Highest Honors (17.43/20)

Languages & Activities

Arabic (Native), English (B2), French (B2)
Fablab ENIT *(2023 – 2025)* – Senior Member
Securinets ENIT *(2025)* – Founding Member
ENIT Junior Enterprise *(2023 – 2025)* – Online mission head
INJAZ Company Program – Econics *(2024)* – Marketing Manager